WebFundamentals

Question 1: What is Web 3.0, and how is it different from Web 1.0 and Web 2.0?

The evolution of the Web can be understood across three generations, each fundamentally changing how users interact with the Internet (Berners-Lee and Fischetti, 1999; Spivack, 2007).

Feature	Web 1.0	Web 2.0	Web 3.0
Era	Early 1990s – early 2000s	Mid-2000s – present	Emerging (2020s onwards)
Nature	Read-only, static	Read-write, interactive	Read-write-own, decentralised
User role	Passive consumer	Active contributor	Owner and participant
Data control	Website owner	Central platforms (Google, Meta)	Individual users (via blockchain)
Technology	Static HTML pages	AJAX, APIs, social platforms	Blockchain, AI, semantic web
Examples	Early corporate websites	Facebook, YouTube, Wikipedia	Cryptocurrency wallets, dApps, NFTs

Web 1.0 consisted of static, read-only pages where content was created exclusively by website owners. Users could only browse; they could not interact, comment, or contribute in any way (Berners-Lee and Fischetti, 1999).

Web 2.0 introduced dynamic, user-generated content and interactivity. Social media platforms, blogs, and wikis allowed users to create and share content. However, data remained centralised in the hands of large technology companies (O'Reilly, 2005).

Web 3.0 is the next evolution of the Web, characterised by decentralisation, artificial intelligence, and user ownership of data. Key features include:

- **Decentralisation:** Data is stored across a distributed network using blockchain technology rather than on centralised servers owned by corporations.
- **Semantic web:** Machines can understand the meaning and context of web content, enabling more intelligent search and data processing.
- **User ownership:** Users own and control their own data, digital assets, and identity — no single company holds authority over it.
- **AI integration:** Artificial intelligence personalises and automates experiences across the web in far more sophisticated ways than Web 2.0 algorithms (Spivack, 2007).

In summary, while Web 1.0 was a passive information network and Web 2.0 was an interactive social network, Web 3.0 is a decentralised, intelligent, and user-owned network where data and digital assets belong to the individual, not a central authority.

Question 2: What are the functional differences between the front end and the back end of a web application?

A web application is divided into two main parts — the front end and the back end — which work together to deliver a complete user experience (Duckett, 2011).

Aspect	Front End	Back End
Also known as	Client-side	Server-side
Runs on	The user's browser (client device)	The web server
Visible to user?	Yes — everything the user sees	No — hidden from the user
Core technologies	HTML, CSS, JavaScript	Python, Node.js, PHP, Java, Ruby
Responsibilities	Layout, styling, interactivity, navigation, forms	Business logic, database queries, authentication, APIs
Examples	Login form, navigation menu, product image gallery	Validating credentials, fetching user data, processing payments

The front end is everything the user sees and interacts with directly in the browser. It is built using three core technologies:

- **HTML** defines the structure and content of the page.
- **CSS** controls the visual appearance — colours, fonts, layout, and responsiveness.
- **JavaScript** adds interactivity — animations, form validation, dynamic content updates without page reloads.

The back end runs on the server and handles everything the user cannot see. It is responsible for:

- Processing HTTP requests sent by the front end.
- Executing business logic (e.g., calculating the total price of a shopping cart).
- Communicating with databases to store and retrieve data.
- Authenticating users and enforcing security rules.
- Sending HTTP responses with the requested data back to the client (Duckett, 2011).

The front end and back end communicate via the HTTP request-response cycle. For example, when a user submits a login form, the front end sends an HTTP POST request containing the username and password to the back end. The back end validates these credentials against the database and sends back a response — either a success (200 OK) with the user's data, or an error (401 Unauthorised) if the credentials are incorrect.

Question 3: Explain the process from typing a URL to viewing the requested page, including the HTTP request-response cycle

When a user types a URL into a browser address bar, a series of steps take place behind the scenes before the requested page is displayed on screen (Kurose and Ross, 2017).

Step-by-step process:

- **1. URL parsing:** The browser reads the URL and identifies its components — the protocol (HTTPS), the domain name (e.g., www.google.com), the path (/search), and any query parameters (?q=web).
- **2. DNS lookup:** The browser contacts a Domain Name System (DNS) server to translate the human-readable domain name into a numeric IP address (e.g., 142.250.80.46). This is similar to looking up a phone number in a directory.
- **3. TCP connection:** The browser establishes a TCP (Transmission Control Protocol) connection

with the server at that IP address on port 443 (for HTTPS) or port 80 (for HTTP).

- **4. TLS handshake (HTTPS only):** If the site uses HTTPS, an encrypted TLS session is established between the client and server to secure all data in transit.
- **5. HTTP request sent:** The browser sends an HTTP GET request to the server, asking for the resource at the specified path.
- **6. Server processes the request:** The server receives the request, executes any back-end logic needed (e.g., fetching data from a database), and prepares a response.
- **7. HTTP response received:** The server sends back an HTTP response containing a status code, response headers, and the requested content in the body (e.g., an HTML file).
- **8. Browser renders the page:** The browser parses the HTML and makes additional requests for linked resources (CSS stylesheets, JavaScript files, images). Once all resources are loaded, the page is rendered and displayed to the user (Kurose and Ross, 2017).

Example HTTP Request Message:

```
GET /index.html HTTP/1.1
Host: www.example.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
Accept: text/html,application/xhtml+xml
Accept-Language: en-ZA,en;q=0.9
Connection: keep-alive
```

In this request: `GET` is the HTTP method, `/index.html` is the path to the requested resource, and `HTTP/1.1` is the protocol version. The header lines provide additional context about the client and the request.

Example HTTP Response Message:

```
HTTP/1.1 200 OK
Date: Wed, 07 May 2026 08:00:00 GMT
Server: Apache/2.4.41 (Ubuntu)
Content-Type: text/html; charset=UTF-8
Content-Length: 6821
Last-Modified: Mon, 05 May 2026 10:15:00 GMT
Connection: close

<!DOCTYPE html>
<html>
  <head><title>Example Page</title></head>
  <body>
    <h1>Welcome to Example.com</h1>
  </body>
</html>
```

In this response: `HTTP/1.1 200 OK` is the status line confirming the request succeeded. The header lines describe the server, content type, and content length. The body contains the actual HTML content that the browser will render for the user.

This entire cycle — from the user pressing Enter to the page appearing on screen — typically completes in milliseconds, demonstrating the efficiency of the HTTP protocol and the client-server architecture that underpins the World Wide Web.

Reference List

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